



Advanced Threat Incident Analytics & Security Visualization Framework

"Intelligent Security. Actionable Insights. Safer Tomorrow."

7

Analytics
Modules

4

Delivery
Milestones

8

Week
Rollout Plan



Problem Statement



THE CHALLENGE

- Security data is scattered across multiple systems
- Teams cannot see the full threat picture easily
- Attacks are not detected on time
- Response is very slow — avg 90 minutes
- No visibility on which area is most at risk
- Future attacks cannot be predicted
- Decision making is slow and reactive



THE SOLUTION

- Collects all attack data in one unified platform
- Shows which attacks are most serious & critical
- Maps threat hotspots across 10 countries
- Tracks how fast security teams respond
- Predicts where next attacks may happen
- One executive dashboard for all decisions
- Real-time situational awareness for all teams

"Too much data — but no clear picture of what is actually happening."

Dataset Overview



1,00,000

Total Records



15

Total Columns



10

Countries



8

Industries



8

Attack Types



2023–24

Time Period



All 15 Columns & Their Purpose

 attack_type	Type — Phishing, DDoS, Ransomware
 target_system	System — Cloud, IoT, Database
 outcome	Result — Success or Failure
 timestamp	When attack happened
 attack_severity	Danger score 1 to 10
 data_compromised_GB	Data stolen in GB
 location	Country of attack

 industry	Sector attacked
 response_time_min	Team response in minutes
 mitigation_method	How attack was stopped
 security_tools_used	Firewall, MFA, Antivirus
 user_role	Admin, Employee, External User
 attack_duration_min	How long attack lasted
 attacker_ip / target_ip	IP addresses of attacker & victim



Data Cleaning Process



Missing Values

Before: 8 cells

After: 0 cells



Duplicate Rows

Before: 3 rows

After: 0 rows



Invalid Values

Before: 4 values

After: 0 values



Data Quality

Before: 68%

After: 100%



6 Data Cleaning Steps



Step 1

Missing Values

Filled empty cells with "Unknown" or "None" using fillna()



Step 2

Invalid Values

Fixed severity > 10, negative response time & negative GB values



Step 3

Duplicate Rows

Removed 3 duplicate rows using drop_duplicates() function



Step 4

Inconsistent Text

Standardized lowercase/UPPERCASE using str.capitalize() & str.strip()



Step 5

Wrong Timestamp Format

Converted all dates to YYYY-MM-DD HH:MM:SS standard format



Step 6

Data Type Mismatch

Converted columns to correct int/float/datetime data types

★ Star Schema Design

We divided our raw dataset into 1 Fact Table + 5 Dimension Tables — organized like a star for fast analysis

DIM_TIME

Hour, Day
Month, Year

Rows: 1,00,000

FACT_INCIDENT (Main Table — 1,00,000 rows)

- TSI Score
- Attack Severity
- Response Time
- Data Compromised GB
- Outcome

DIM_THREAT

Attack Type
Severity Level

Rows: 8

DIM_LOCATION

Country
Region

Rows: 10

DIM_RESPONSE

Response Time
Mitigation Method

DIM_INDUSTRY

Industry
Target System

Rows: 8

⚡ TSI Score — Threat Severity Index

TSI tells us how dangerous each attack is — combining 3 key security factors

$$\text{TSI} = (\text{Attack Severity} \times 0.4) + (\text{Data Compromised} / 100 \times 10 \times 0.3) + (\text{Response Time} / 180 \times 10 \times 0.3)$$



Attack Severity

40% Weight

Shows how dangerous the attack is

Example:
Severity = 7
 $7 \times 0.4 = 2.8$



Data Compromised

30% Weight

Shows how much damage was done

Example:
Data = 65 GB
 $(65/100 \times 10) \times 0.3 = 1.95$



Response Time

30% Weight

Shows how late team responded

Example:
Time = 90 min
 $(90/180 \times 10) \times 0.3 = 1.5$

TSI Score Ranges:

0–3 Low

3–5 Medium

5–7 High

7–10 Critical

Our Dataset TSI Score = 5.21 → HIGH THREAT Level | Ransomware is the only Critical level attack



Dashboard — Page 1: Overview

Milestone 1 | Weeks 1–2 | Data Integration & Incident Modeling

 Total Incidents

1,00,000

 Avg TSI Score

5.21

 Avg Response Time

90 Min

 Data Compromised

5.01M GB



Incidents by Attack Type

Chart: Horizontal Bar Chart

Why: Best for comparing 8 attack type categories side by side



Threat Severity Distribution

Chart: Donut Chart

Why: Shows Critical/High/Medium/Low split of all incidents



Attack Outcome

Chart: Donut Chart

Why: Shows Success vs Failure proportion out of total incidents



Most Targeted Industries

Chart: Column Chart

Why: Compares all 8 industries — which is most targeted



Dashboard — Page 2: Threat Intelligence

Milestone 2 | Weeks 3–4 | Threat Intelligence & Temporal Analytics

⚡ Overall TSI Score

5.21

HIGH Threat

🔴 Critical Incidents

12,499

Ransomware

📊 High Severity Count

25,024

Zero-Day & Malware

🧠 Avg Severity Score

5.50

Out of 10

⚡ TSI Gauge Chart

Chart: Gauge Visual

Why: Shows 5.21 score with needle — High vs Critical zone

🌀 Incident Categories

Chart: Donut Chart

Why: 8 attack types — equal ~12.5% distribution each

📊 Monthly Incident Trend

Chart: Column Chart

Why: Monthly patterns — March highest 8.6K, Feb lowest 7.9K

🔧 Security Tools Usage

Chart: Bar Chart

Why: Endpoint Detection most used with 1,45,231 deployments



Dashboard — Page 3: Geospatial & Response

Milestone 3 | Weeks 5–6 | Geospatial Intelligence & Response Analytics

 Most Attacked Country

India

Highest response time

 Avg Response Time

90.78 Min

Target: 60 min

 Total Locations

10

Countries attacked

 Max Response Time

180 Min

Slowest recorded

Global Threat Hotspot Map

Chart: Map Visual (Bing Maps)

Why: Geographic data needs map — bubbles show attack concentration by country

Industry vs Location Heatmap

Chart: Matrix Visual

Why: Shows which industry in which country is most attacked — color coded

Avg Response Time by Location

Chart: Horizontal Bar Chart

Why: Compare response time across 10 countries — India slowest at 91.3 min

Key Insights Box

Chart: Text Box + Anomaly Alert

Why: Summarizes top 3 findings from geospatial analysis for quick reading



Dashboard — Page 4: Executive Summary

Milestone 4 | Weeks 7–8 | Risk Intelligence & Executive Analytics

⚡ Overall TSI

5.21

📄 SLA Compliance

33.3%

✅ Attacks Resolved

~50,030

💾 Data at Risk

5.01M GB

🚨 Critical Attacks

12,499

⌚ Avg Duration

151.8 Min



Incident Trend & Forecast

Chart: Line Chart (Dual Axis)

Why: Shows monthly trend + TSI score together for executive overview



Industry Risk Analysis

Chart: Horizontal Bar Chart

Why: Compares all 8 industries — Government highest at 12,645



Security Risk Scorecard

Chart: Table Visual

Why: Industry-wise risk classification with conditional color formatting



Platform Risk Gauge

Chart: Gauge Visual

Why: Shows overall TSI at 5.21 vs target 5.0 with SLA & uptime stats



Key Findings & Insights

 Most Common Attack

Brute Force

12,605 incidents — highest among all 8 attack types

 Most Attacked Country

India

91.3 min response time — highest across all 10 countries

 Most Used Security Tool

Endpoint Detection

1,45,231 deployments — first line of defense

 Platform TSI Score

5.21 — HIGH

High threat level — immediate security improvements needed

 Most Targeted Industry

Government

12,645 incidents — finance and manufacturing follow closely

 Highest Attack Month

March

8,600 incidents — followed by August and December at 8,500

 Attack Success Rate

50% vs 50%

50,030 successful — 49,970 stopped by security teams

 Total Data Compromised

5.01 Million GB

Massive scale of damage across all incidents and industries



8-Week Delivery Roadmap

Milestone 1

Week 1–2

Data Integration & Incident Modeling

- ✓ Collected & cleaned dataset
- ✓ Built ETL pipeline with Pandas
- ✓ Created Star Schema — 6 tables
- ✓ Calculated TSI Score for 1L records

✓ COMPLETE

Milestone 2

Week 3–4

Threat Intelligence & Temporal Analytics

- ✓ Built TSI Gauge dashboard
- ✓ Monthly trend analysis
- ✓ Incident categories classified
- ✓ Security tools usage analyzed

✓ COMPLETE

Milestone 3

Week 5–6

Geospatial Intelligence & Response Analytics

- ✓ World threat hotspot map
- ✓ Country-wise response time
- ✓ Industry vs Location heatmap
- ✓ Response bar chart built

✓ COMPLETE

Milestone 4

Week 7–8

Risk Intelligence & Executive Analytics

- ✓ Security Risk Scorecard
- ✓ Industry risk analysis bar
- ✓ Platform risk gauge built
- ✓ Executive command center done

✓ COMPLETE



Tools & Technologies Used



Python

Core programming language for all data processing tasks



Pandas

Data cleaning, ETL pipeline, Star Schema table creation



Jupyter Notebook

Development environment for running Python code



Power BI Desktop

Building all 4 interactive dashboard pages



Star Schema

Data warehouse design — 1 Fact + 5 Dimension tables



GitHub

Version control and project hosting platform



Matplotlib/Seaborn

Python visualization libraries for EDA



Power BI Maps

Geospatial visualization using Bing Maps integration



Thank You!

Advanced Threat Incident Analytics & Security Visualization Framework



1,00,000
Records



Star Schema
6 Tables



4 Page
Dashboard



TSI Score
Calculated



10 Countries
Mapped



4 Milestones
Complete

Infosys Springboard Virtual Internship 7.0 | Batch 2 | Data Visualization

"Intelligent Security. Actionable Insights. Safer Tomorrow."



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